

Lab Problem 1

Objective:

To implement multivariate linear regression and analyze the convergence of the model using different optimization strategies. You will visualize the cost function, tune the learning rate (α), and compare the behavior of **Batch Gradient Descent (BGD)** versus **Stochastic Gradient Descent (SGD)**.

Dataset Used:

The California Housing Dataset (available via `sklearn.datasets.fetch_california_housing`).

Task:

1. **Data Preparation:** Load the dataset and split it into a training set (80%) and a testing set (20%).
2. **Cost Function Visualization:** Implement a routine to track the value of the cost function (Mean Squared Error) at each iteration during the training process.
 - a. **Task:** Generate a plot showing **Cost vs. Number of Iterations**.
3. **Learning Rate Optimization:** Experiment with at least three different values for the learning rate parameter (α) (e.g., 0.0001, 0.001, 0.01).
 - a. **Task:** Observe how the choice of α impacts the speed of convergence and whether the cost function oscillates or diverges.
4. **Optimization Comparison:** Implement and compare the training process using the following two methods:
 - a. **Batch Gradient Descent (BGD):** Updating weights after calculating the gradient over the entire training set.
 - b. **Stochastic Gradient Descent (SGD):** Updating weights after each individual training sample.
 - c. **Task:** Compare the two methods in terms of computational time and the "smoothness" of the cost function reduction in your plots.
5. **Final Evaluation:** Using the optimized parameters from your best-performing gradient descent run, calculate the final **Mean Squared Error (MSE)** on the test set.